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The Technology Value Stream

Lead time, scenarios, and the DevOps ideal

Source: The DevOps Handbook, Chapter 1

Value Stream Basics

- Sequence of activities delivering customer value
- Manufacturing: order received, materials released
- Technology: idea becomes running service
- Value exists only in production
- Our focus: deployment lead time

(Kim et al., 2021; Martin & Osterling, 2013)

Lead Time vs. Processing Time

Lead Time

- Clock starts at the request
- Clock ends when fulfilled
- Includes time waiting in queues
- What the customer experiences

Processing Time

- Clock starts when work begins
- Omits time spent in queues
- Also called touch time
- Measures hands-on effort only

(Kim et al., 2021, p. 10)

Lead Time on a Timeline

Request made — Request fulfilled

Lead time: the whole bar



Processing time: blue segments only

Illustration based on (Kim et al., 2021, p. 10)

The Common Scenario

Deployment lead times requiring months

- Large, complex organizations
- Tightly coupled, monolithic systems
- Scarce integration test environments
- Long test and production lead times
- Heavy reliance on manual testing
- Multiple required approval processes

(Kim et al., 2021, p. 10)

Months in Practice

3
months
Figure 1.2 lead time

- Heroics needed at nearly every stage
- Merged code may fail every test
- Fixes take days or weeks
- Customer outcomes remain poor

(Kim et al., 2021, p. 10; Edwards, 2015)

Our DevOps Ideal

Deployment lead times of minutes

- Fast, constant developer feedback
- Small changes checked in continually
- Automated and exploratory testing
- Modular, loosely coupled architecture
- Small teams work autonomously
- Failures stay small and contained

(Kim et al., 2021, p. 10)

Months Versus Minutes

	Common Scenario	DevOps Ideal
Lead time	Months	Minutes, or hours
Architecture	Monolithic, tightly coupled	Modular, loosely coupled
Testing	Manual and late	Automated and continuous
Failures	Global disruptions	Small and contained
Effort	Constant heroics	Fast, steady feedback

(Kim et al., 2021, p. 10)

Key Takeaways

- Lead time is the customer’s experience
- Waiting in queues drives long lead times
- Months-long delivery demands constant heroics
- Small batches enable deployment in minutes
- Architecture and automation enable fast flow

References

Edwards, D. (2015). *DevOps Kaizen* [Figure 1.2 source]. In Kim et al. (2016).

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